TASK 7 | ISPU IMPLEMENTATION OF QUAT+FE

Implementation evidence, model validation, communication, memory, and benchmark status

Status: demonstrated on LSM6DSO16IS ISPU and STM32F401RE; observed activity IDs 0 (Horizontal), 1 (Static), and 2 (Vertical). Quantitative ISPU timing and total power values still require hardware measurements. This appendix is designed to accompany the existing Tasks 1–6 report, not replace it.

1. System and implementation

Subsystem	Implementation / evidence
Hardware	NUCLEO-F401RE + X-NUCLEO-IKS4A1; LSM6DSO16IS with ISPU
IMU / ISPU	±8 g acceleration; ±2000 dps gyro; native sensor/ISPU cycle 208 Hz
Quaternion fusion	Custom calibration-aware six-axis Madgwick C on ISPU; output w,x,y,z (displayed x,y,z,w)
Quaternion export	STM32 AlgoBuilder timer at 120 Hz; measured ~120.48–120.51 Hz in recorded sessions
Classifier	42-sample windows, 21-sample stride; eight variance/peak-to-peak quaternion features; 17-node depth-4 tree
Classes	0 Horizontal; 1 Static; 2 Vertical; 255 not ready
Communication	STM32F401RE polls 64 bytes of ISPU DOUT registers; USART2 921600 8-N-1 binary protocol to MEMS Studio

Architecture: IMU → Madgwick on ISPU → quaternion window & QUAT+FE decision tree on ISPU → 64-byte ISPU DOUT registers → STM32 timer-driven polling → MEMS Studio / UART. The STM32 does not perform Task 7 decision-tree inference.

2. Training and evaluation

Class	120 Hz samples	Duration (s)	Rate (Hz)
Horizontal	3655	30.321	120.51
Static	3653	30.312	120.48
Vertical	3659	30.361	120.48

The original appended log files contained earlier ~100 Hz runs. Only the later ~120 Hz sessions were extracted for training. A chronological per-recording split (60% train, 15% validation, 15% test, with gap segments) was used. This is a within-session evaluation, not independent repeated-trial validation.

Metric	Motion-feature tree	RFE comparison tree
Train accuracy	288/307 (93.81%)	290/307 (94.46%)
Validation accuracy	75/78 (96.15%)	78/78 (100%)
Test accuracy	75/78 (96.15%)	53/78 (67.95%)
Tree nodes / depth / leaves	17 / 4 / 9	7 / 2 / 4

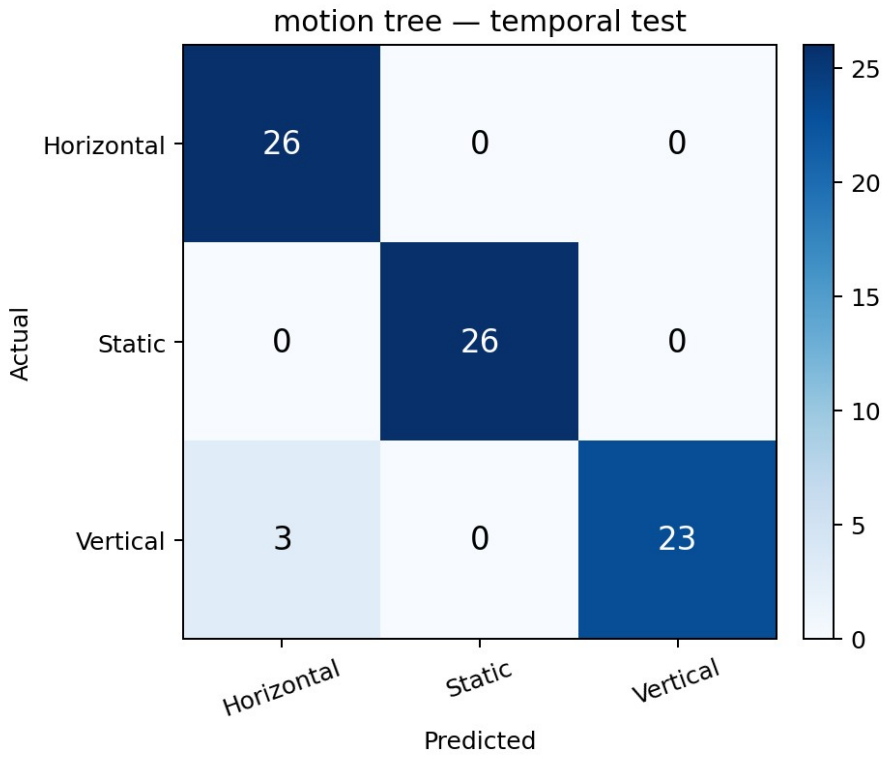


Figure T7-1. Chronological test confusion matrix; axes use model classes 0=Horizontal, 1=Static, 2=Vertical.

The test confusion matrix (true rows, predicted columns) is [[26,0,0],[0,26,0],[3,0,23]]: 75 of 78 windows classified correctly. Because different portions of the same activity recording appear in the splits, these results should not be interpreted as independent hardware accuracy.

3. Python/C reference validation

The bundled desktop checker reproduced the Python/JSON tree output using the generated C feature-extraction and traversal routines on 463 windows: 463/463 matched (100% agreement for that offline check). This establishes numerical equivalence of those implementations on the checked windows, not alignment with predictions captured from separate live ISPU trials.

The tree outputs a leaf-training-purity field as a heuristic quality indicator; it is NOT a calibrated probability of correctness. Fresh ground-truth-labeled hardware trials are still required for a live ISPU-vs-Python confusion matrix and agreement estimate.

4. Source files and compact tree representation

The accompanying package includes native C source and headers: ISPU_208hz/src/main.c, madgwick.c/.h, fusion_core.c/.h, and quat_fe_tree.c/.h, plus the STRED Makefile and sensor configuration. The decision-tree header defines QuatFeNode with feature, left/right children, threshold, predicted class and leaf training purity, and the C array has 17 nodes. Model JSON and training scripts are included for reproducibility.

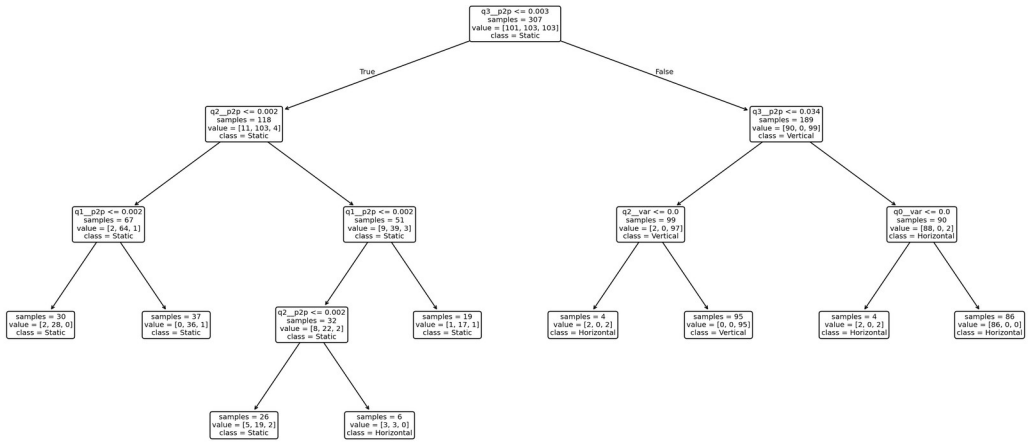


Figure T7-2. Decision-tree structure generated from the 120 Hz QUAT+FE dataset.

5. STM32F401RE reception and UART

The generated STM32 `algo_builder.c` calls Sensor_Hub_Init(0,120,2,3) and ISPU_Init(64). The Sensor_Hub_Handler path in `ab_sensor_hub.c` selects the ISPU register bank and reads all 64 DOUT bytes through the IKS4A1 BSP register API (timer-driven polling). The activity-class ID is extracted at byte offset 40; the UART/display layer sends the selected values to MEMS Studio.

In `com.c`, `Usart_BaudRate = 921600`; `USARTConfig()` configures 8 data bits, 1 stop bit, no parity and no hardware flow control. Actual packets use HAL_UART_Transmit with framed/binary MEMS Studio protocol, not plain-text terminal messages. The corresponding code evidence is bundled.

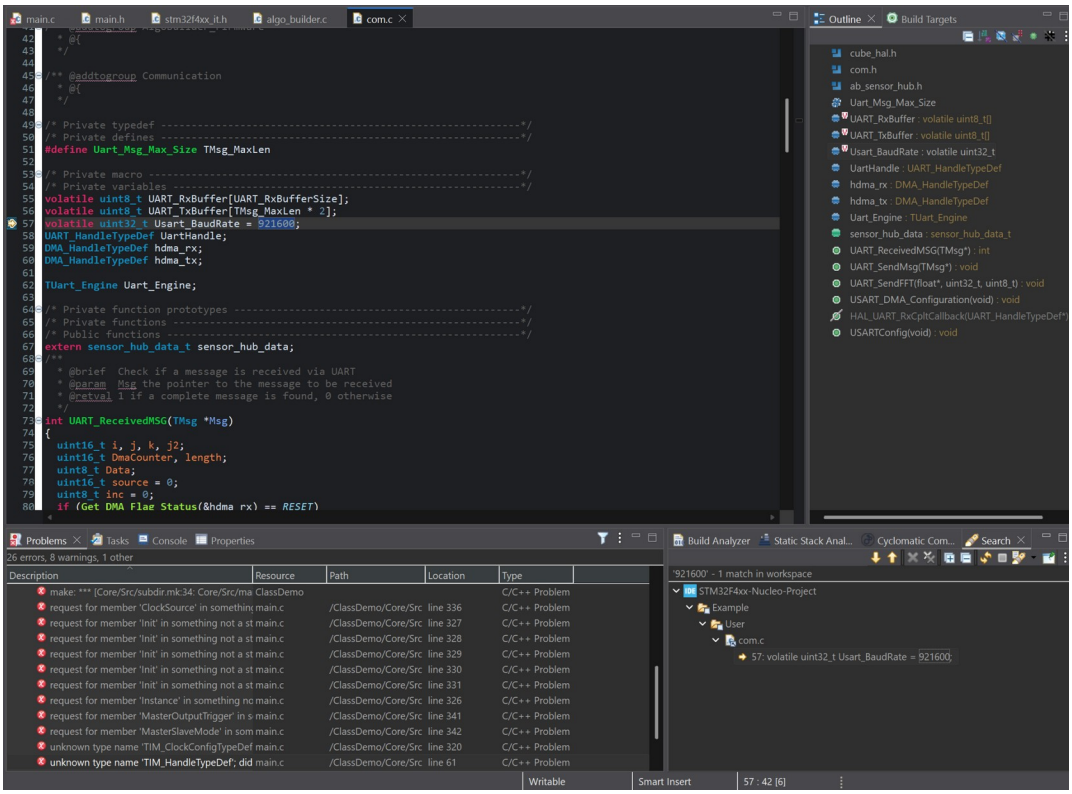


Figure T7-3. STM32CubeIDE code showing the configured 921600 baud value.

6. ISPU memory utilization

Section	Bytes	KiB	Meaning
text	21932	21.418	~66.93% of 32 KiB ISPU program memory
data	108	0.105	Initialized variable section
bss	784	0.766	Uninitialized variable section
data + bss	892	0.871	~10.89% of 8 KiB ISPU data memory
dec	22824	22.289	Sum of ELF sections, not one bank

These are the final STRED `stred-size` figures reported from the successful ISPU build. They exclude complete runtime stack/high-water usage. STM32 FLASH/RAM overhead must be extracted from the STM32 ELF/MAP after building the actual generated project; it is not represented by this ISPU size output.

7. Benchmark table: measured vs. outstanding

Quantity	ISPU implementation	STM32-only reference
Total processing time per sample	NOT MEASURED: INT2 active pulse needed	NOT MEASURED
Decision-tree-only traversal time	NOT ISOLATED	NOT MEASURED
CPU cycles / sample	NOT MEASURED: requires timing and ISPU clock	NOT MEASURED: DWT required
Program/data memory	21.418 / 0.871 KiB	NOT MEASURED: MCU ELF/MAP
Power	Sensor-only baseline below; total unavailable	NOT MEASURED

Power baseline: ST DS13892 specifies approximately 595 µA for combined accelerometer and gyroscope high-performance operation at VDD=1.8 V (ISPU current excluded). Therefore, sensor-only typical baseline power is ~1.071 mW. This is not the measured or estimated total power of the running application. Do not compare this baseline with an STM32-only full-system number.

For total ISPU active time, AN5799 §9.9 describes routing the ISPU sleep indication to INT2; its inactive pulse duration reflects processing time. A benchmark-only conf.txt is supplied. Record measured LOW-pulse widths, ISPU clock and current assumptions in the benchmark worksheet; then compute cycles and duty-cycle-based power. Run an STM32-only reference with DWT instrumentation for an apples-to-apples comparison.

8. Real-time hardware demonstration and limitations

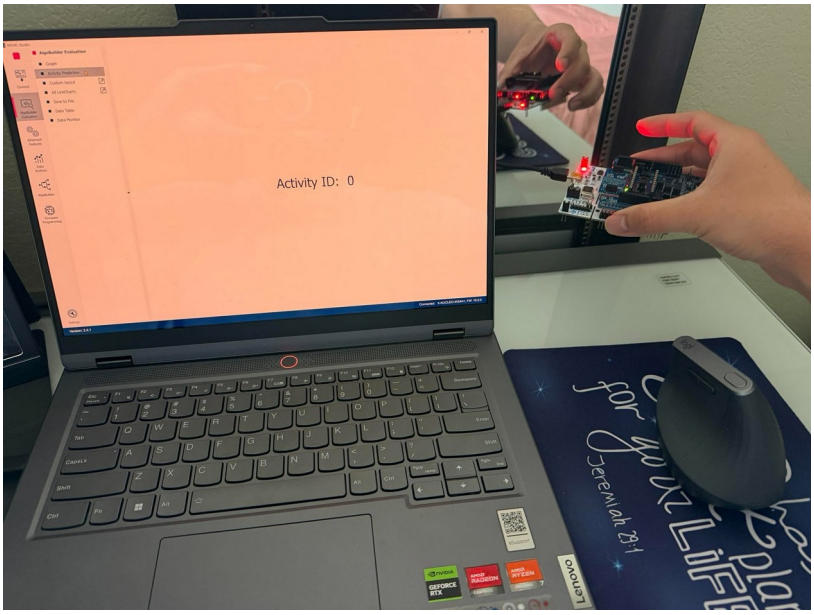


Figure T7-4. Representative live MEMS Studio classification with the physical NUCLEO/IKS4A1 board (Activity ID 0).

Live demonstrations were reported successful for classes 0, 1 and 2; photographs and graph screenshots are included in folder 06_Hardware_Evidence. Observing changing class IDs verifies execution and communication, but not a quantified out-of-sample hardware confusion matrix.

9. Submission status and remaining evidence

Expected output	Status
ISPU C source; compact QUAT+FE model header	INCLUDED
JSON training model; training/validation plots	INCLUDED
Working MEMS Studio native .mems / final plsworkv1.json	NOT AVAILABLE IN PROVIDED FILES — export working design
STM32 polling and UART integration source evidence	INCLUDED; full generated buildable workspace not provided
ISPU and STM32 memory report	ISPU INCLUDED; STM32 memory PENDING
ISPU vs STM32 latency / CPU cycles / power	WORKSHEET INCLUDED; hardware measurement PENDING
Python vs C offline agreement	463/463 INCLUDED
Hardware-vs-reference confusion matrix	PENDING independent labeled sessions
Live screenshots	INCLUDED

References

[1] STMicroelectronics, LSM6DSO16IS datasheet DS13892, Rev. 2, June 2022, https://www.st.com/resource/en/datasheet/lsm6dso16is.pdf

[2] STMicroelectronics, AN5799, LSM6DSO16IS application note, Section 9.9 (INT2 active-time monitoring), https://www.st.com/resource/en/application_note/an5799-lsm6dso16is-alwayson-3axis-accelerometer-and-3axis-gyroscope-with-ispu--intelligent-sensor-processing-unit-stmicroelectronics.pdf

[3] User-provided MEMS Studio 2.4.1 generated source: algo_builder.c, ab_sensor_hub.c, and com.c; final STRED `stred-size` output.

[4] Task 7 model development artifacts: retrain_task7.py, evaluation_results.json, tests/verify_c_matches_python.py, host_validation_output.txt.